Mengjie Zhang

Biomechanics and Ergonomics · Software Development

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• Auckland, New Zealand

Scan for portfolio and system demos

10+	3	4	2
Years in sport science and biomechanics	Oral presentations in 2026	Assessment modules in one video-based web application	First-author journal articles

Education

2023 – 2027 (expected)
PhD, Biomechanics and Ergonomics
The University of Auckland, New Zealand

2020 – 2023
Master's degree, Sport Human Science
Beijing Sport University, China

2016 – 2020
Bachelor's degree, Sport Human Science
Beijing Sport University, China

Technical skills

HUMAN MOVEMENT ANALYSIS
Vicon · Motion Analysis (Cortex) · OpenCap · MediaPipe · OpenCV

BIOMECHANICS
Inverse dynamics · OpenSim · Visual3D · Force platforms · Lumbar loading

ERGONOMICS
RULA · REBA · Revised NIOSH · 3DSSPP · Manual handling

SOFTWARE AND AI
Python · Streamlit · PyTorch · pandas · Plotly · MATLAB

RESEARCH AND TESTING
Validation studies · SPSS · Gas analysis · VO_2max · Isokinetics · sEMG

Scholarship

China Scholarship Council (CSC) doctoral scholarship, Oct 2023 – Oct 2027 (48 months)

Languages

Chinese (native) · English (professional working proficiency)

Profile

Developing validated software, grounded in biomechanics research, for human movement analysis and ergonomic design. Building on a background in sport science and sports biomechanics, my work focuses on digital human modelling, human movement analysis, occupational biomechanics and ergonomic design. Doctoral candidate at the University of Auckland, nearing thesis submission, with the PhD expected to be conferred in 2027.

Experience

Research Assistant — Jun 2026 – present

The University of Auckland, New Zealand · PI: Associate Professor Yanxin Zhang

Lead developer of a markerless system for human movement analysis from ordinary single-camera video, requiring no markers, force plates or laboratory set-up.

Gait analysis
Walking kinematics, joint moments and lower-limb muscle activation

Manual-handling risk
Lifting and handling tasks, including lumbar loading estimation

Rehabilitation assessment
Preliminary screening and progress tracking

- Adapted the markerless pose model, including added neck degrees of freedom, and integrated it with inverse dynamics, muscle-activation estimation and a web interface.
- Fine-tuned a deep-learning model that predicts ground reaction forces from video-derived motion, which substantially reduced vertical ground reaction force error in lifting tasks.
- Automated exports of the analysis video, PDF report and joint-angle data.

Doctoral Researcher — 2023 – present

The University of Auckland, New Zealand

- Designed and built a four-module web application that converts workplace video into postural risk, manual-handling and biomechanical outputs with automated reports.
- Evaluated against marker-based motion capture and 3DSSPP in laboratory lifting trials: postural risk categories at the peak posture agreed closely with the reference, and four experts rated the system useful and easy to use.
- Led a concurrent-validity study of markerless motion capture against a Vicon reference in occupational lifting (Applied Ergonomics, 2027).
- Developed a hybrid two-phase posture prediction framework for design-stage ergonomic evaluation, validated against marker-based motion capture and fast enough for interactive use (WCB 2026).
- Presented at the 10th World Congress of Biomechanics and the HFESNZ 40th Conference (2026).

Experience

Core Member, Footwear Performance Testing
Mar – Oct 2022

Beijing Sport University, China

- Industry-commissioned running-economy testing of a racing-shoe series: randomised crossover study with 15 male recreational runners covering maximal oxygen uptake, running economy and lower-limb biomechanics.
- Collected data with an 8-camera infrared motion capture system (200 Hz) and a Kistler force platform (1,000 Hz); analysed outcomes with two-way repeated-measures ANOVA.

Project Lead, Master's Research
Nov 2021 – Jun 2022

Beijing Sport University, China

- Designed and led a repeated-measures study of three commercially available racing shoes with different midsole bending stiffness, measuring running economy, metabolic energy cost and lower-limb joint kinematics and kinetics.
- Presented at the 40th Conference of the International Society of Biomechanics in Sports (2022).

Intern, Sports Department
2021

Tsinghua University, China

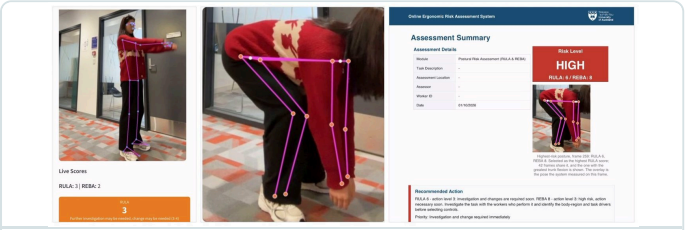
- Physical fitness testing for physical education courses, including the Functional Movement Screen and body composition assessment.

Intern, Physical Fitness Testing for People with Disabilities
2021

Research project, Wuhan, China

- Fitness assessments including anthropometry, blood pressure, lung capacity, strength and speed.

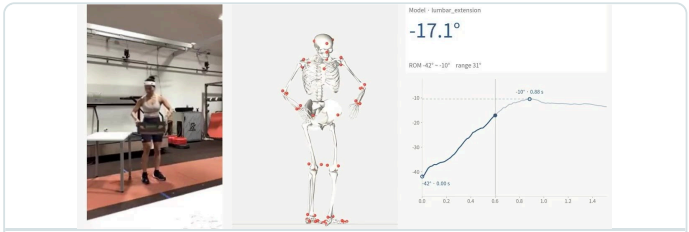
Selected projects



Online video-based ergonomic assessment system

Four modules from a single standard video: RULA and REBA postural risk, the Revised NIOSH Lifting Equation, static 3D inverse dynamics and dynamic biomechanics.

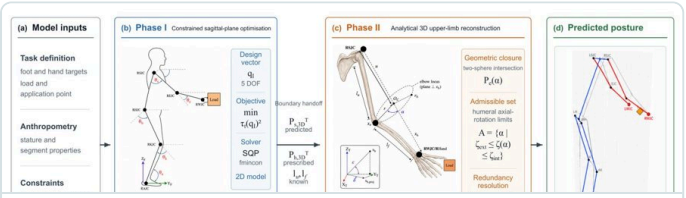
Validated against motion capture **Expert-evaluated**



Single-camera musculoskeletal analysis

From ordinary video to 3D kinematics, predicted ground reaction forces, inverse dynamics and lower-limb muscle activation.

Deep-learning GRF prediction **Web interface**



Hybrid two-phase posture prediction

Predicts working posture from task specifications before a workstation exists, for design-stage ergonomic evaluation.

Interactive speed **WCB 2026**



Footwear performance testing

Running economy and lower-limb biomechanics of racing shoes with different midsole bending stiffness.

15 runners **ISBS 2022**

Selected publications and presentations

[1] **Zhang, M.**, Nieuwenhuys, A., Boocock, M.*, & Zhang, Y.* (2027). Concurrent validity of markerless motion capture for occupational lifting: Kinematics and ergonomic risk assessment. *Applied Ergonomics*, 138, 104866.

[2] **Zhang, M.**, Nieuwenhuys, A., & Zhang, Y.* (2025). Posture prediction models in digital human modeling for ergonomic design: A systematic review. *Medical Engineering & Physics*, 143, 104391.

[3] Oral presentations: 10th World Congress of Biomechanics, Vancouver (2 presentations, 2026) · HFESNZ 40th Conference, Wellington (2026) · 2nd Human Sports Science Conference (2022).

* Corresponding author.

Volunteering

Beijing Marathon (2017) · Beijing Tongzhou Marathon (2021) · Beijing Sport University COVID-19 vaccination programme (2021)